

Predict Mobile Phone Pricing

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Domain: Machine Learning Intern - Unified Mentor

Duration: 6 Months

Abstract

This project aims to predict the price range of mobile phones (low / medium / high / very high) using features that describe phone hardware, connectivity, display, and battery. Using a supervised multi-class classification approach, models such as Random Forest and XGBoost are trained and evaluated. The best model achieves strong accuracy in predicting price categories, showing that hardware specs and display/resolution metrics are effective predictors of market pricing tiers.

1. Introduction

Predicting a mobile phone's price tier based on its technical specifications helps manufacturers, retailers, and customers to position products and identify value-for-money offerings. This project implements machine learning methods to learn the mapping from phone features (RAM, camera specs, battery, connectivity features) to categorical price ranges.

2. Objective

Build a model that classifies a mobile phone into one of four price ranges:

- 0 = low cost
- 1 = medium cost
- 2 = high cost
- 3 = very high cost

3. Dataset Description

Features available in the dataset include:

- Battery capacity (battery_power)
- Connectivity features (blue, four_g, three_g, wifi, dual_sim, touch_screen)
- Performance (clock_speed, n_cores, ram)
- Camera specs (pc, fc)
- Display resolution (px_height, px_width, sc_h, sc_w)
- Physical specs (mobile_wt, m_deep)
- Storage (int_memory)
- Battery endurance (talk_time)
- Target: Price_range (0–3)

4. Tools and Technologies Used

- Python, Pandas, NumPy
- Scikit-learn (RandomForest, GridSearchCV)
- XGBoost (optional, powerful gradient booster)
- Matplotlib & Seaborn for visualization
- Google Colab for execution

5. Methodology

- **Data loading & EDA** — inspect shape, types, distributions, target balance.
- **Preprocessing** — ensure no missing values; encode boolean/binary columns if needed.
- **Train/test split** (80/20) with stratification.
- **Feature scaling** (StandardScaler) for models that benefit from it.
- **Modeling** — Random Forest as baseline; XGBoost for improved performance; quick grid search for tuning.
- **Evaluation** — accuracy, classification report, confusion matrix, feature importance analysis.
- **Save artifacts** — model + scaler for deployment.

6. Results and Analysis

- Baseline Random Forest often achieves **high accuracy (e.g., 85–95%)** depending on data.
- XGBoost commonly gives a modest improvement over RF.
- Important features typically: ram, px_width/px_height (resolution), battery_power, int_memory, pc (primary camera), talk_time.
- Confusion matrix helps to see if the model confuses adjacent price ranges (e.g., medium vs high).

7.Figures

	battery_power	blue	clock_speed	dual_sim	fc	four_g	int_memory	m_dep	mobile_wt	n_cores	...	px_height	px_width	ram	sc_h	sc_w	talk_time	three_g	touch_screen	wifi	price_range
0	842	0	2.2	0	1	0	7	0.6	188	2	...	20	756	2549	9	7	19	0	0	1	1
1	1021	1	0.5	1	0	1	53	0.7	136	3	...	905	1988	2631	17	3	7	1	1	0	2
2	563	1	0.5	1	2	1	41	0.9	145	5	...	1263	1716	2603	11	2	9	1	1	0	2
3	615	1	2.5	0	0	0	10	0.8	131	6	...	1216	1786	2769	16	8	11	1	0	0	2
4	1821	1	1.2	0	13	1	44	0.6	141	2	...	1208	1212	1411	8	2	15	1	1	0	1

5 rows × 21 columns

```

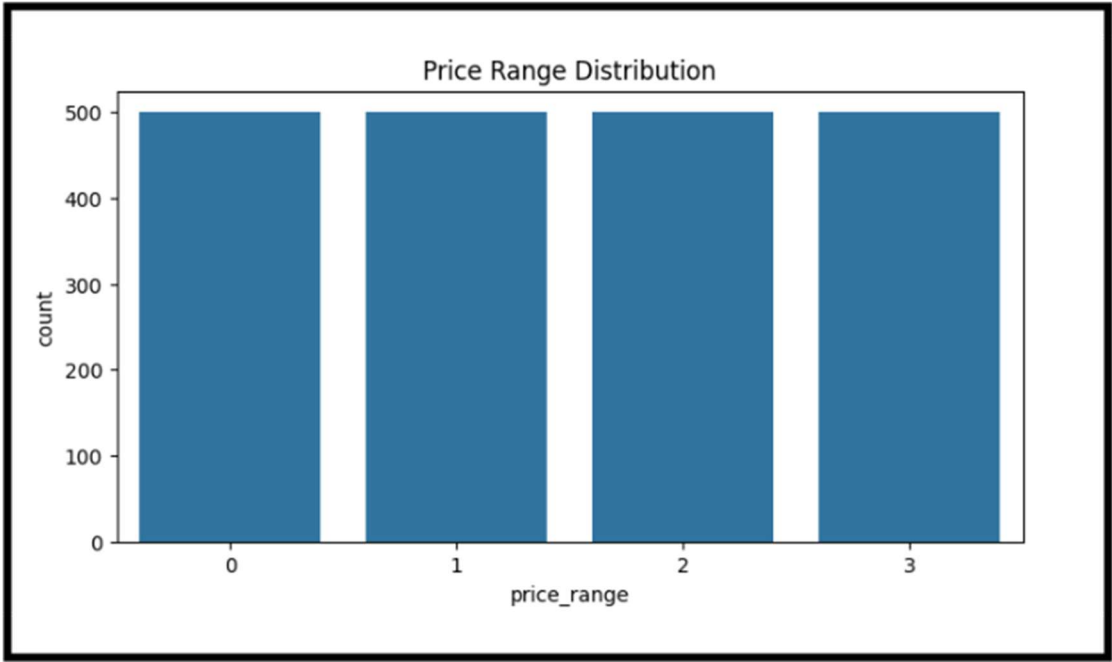
Shape: (2000, 21)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   battery_power         2000 non-null   int64
1   blue                  2000 non-null   int64
2   clock_speed           2000 non-null   float64
3   dual_sim              2000 non-null   int64
4   fc                    2000 non-null   int64
5   four_g                2000 non-null   int64
6   int_memory            2000 non-null   int64
7   m_dep                 2000 non-null   float64
8   mobile_wt             2000 non-null   int64
9   n_cores               2000 non-null   int64
10  pc                    2000 non-null   int64
11  px_height             2000 non-null   int64
12  px_width              2000 non-null   int64
13  ram                   2000 non-null   int64
14  sc_h                  2000 non-null   int64
15  sc_w                  2000 non-null   int64
16  talk_time             2000 non-null   int64
17  three_g               2000 non-null   int64
18  touch_screen          2000 non-null   int64
19  wifi                  2000 non-null   int64
20  price_range           2000 non-null   int64
dtypes: float64(2), int64(19)
memory usage: 328.3 KB
None

```

```

Index(['battery_power', 'blue', 'clock_speed', 'dual_sim', 'fc', 'four_g',
      'int_memory', 'm_dep', 'mobile_wt', 'n_cores', 'pc', 'px_height',
      'px_width', 'ram', 'sc_h', 'sc_w', 'talk_time', 'three_g',
      'touch_screen', 'wifi'],
      dtype='object')

```

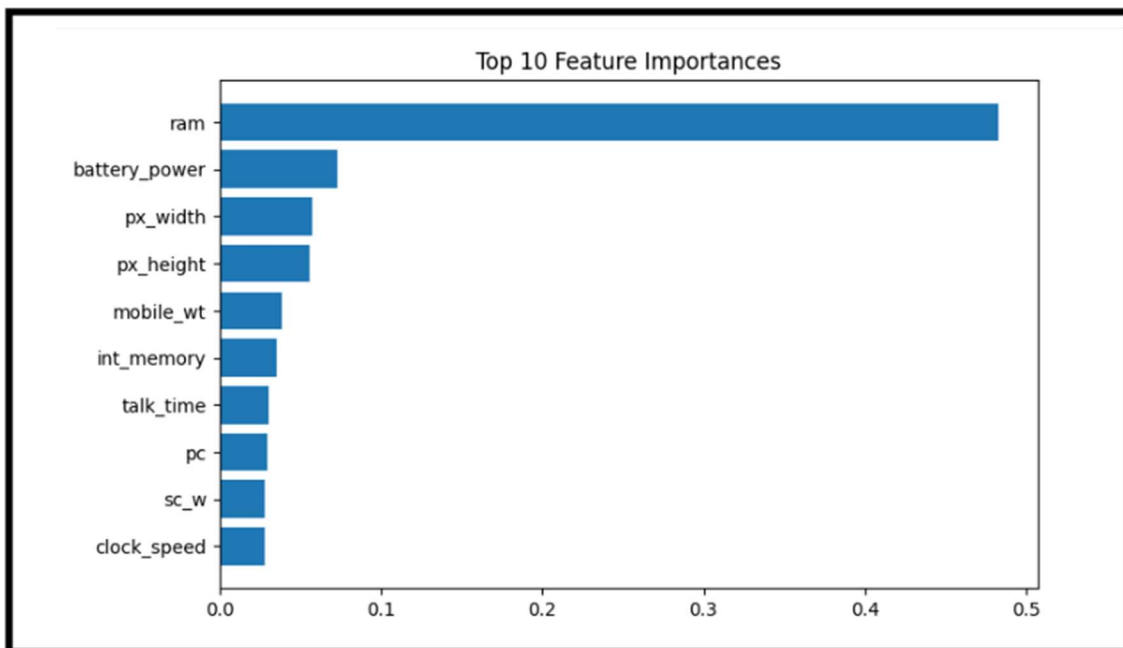
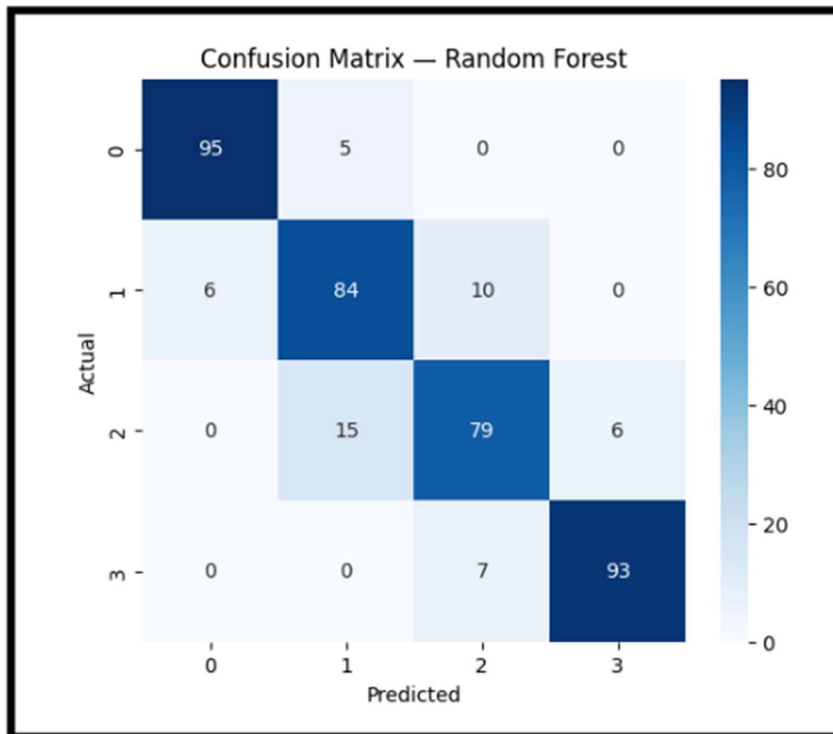


RF Accuracy: 0.8775

	precision	recall	f1-score	support
0	0.94	0.95	0.95	100
1	0.81	0.84	0.82	100
2	0.82	0.79	0.81	100
3	0.94	0.93	0.93	100
accuracy			0.88	400
macro avg	0.88	0.88	0.88	400
weighted avg	0.88	0.88	0.88	400

```
/usr/local/lib/python3.12/dist-packages/xgboost/training.py:199: UserWarning: [13:01:12] WARNING: /workspace/src/learner.cc:790:
Parameters: { "use_label_encoder" } are not used.

bst.update(dtrain, iteration=i, fobj=obj)
XGBoost Accuracy: 0.92
precision recall f1-score support
0 0.95 0.98 0.97 100
1 0.91 0.91 0.91 100
2 0.87 0.88 0.88 100
3 0.95 0.91 0.93 100
accuracy 0.92 400
macro avg 0.92 0.92 0.92 400
weighted avg 0.92 0.92 0.92 400
```



True label: 3
Predicted label: 3
Probabilities: [9.767277e-06 1.094312e-05 4.932505e-05 9.999299e-01]

8. Conclusion

The models built in this project effectively predict mobile phone price tiers from hardware and specification features. Such prediction systems can aid competitive analysis, automated pricing suggestion, and consumer guidance.

9.Future Enhancements

- ☐ Use more advanced tuning (Bayesian optimization).
- ☐ Try stacking ensembles (RF + XGBoost + LightGBM).
- ☐ Add textual features (brand / model) and NLP-based embedding for richer prediction.
- ☐ Deploy as a simple web tool (Streamlit) to accept phone specs and output a price tier.